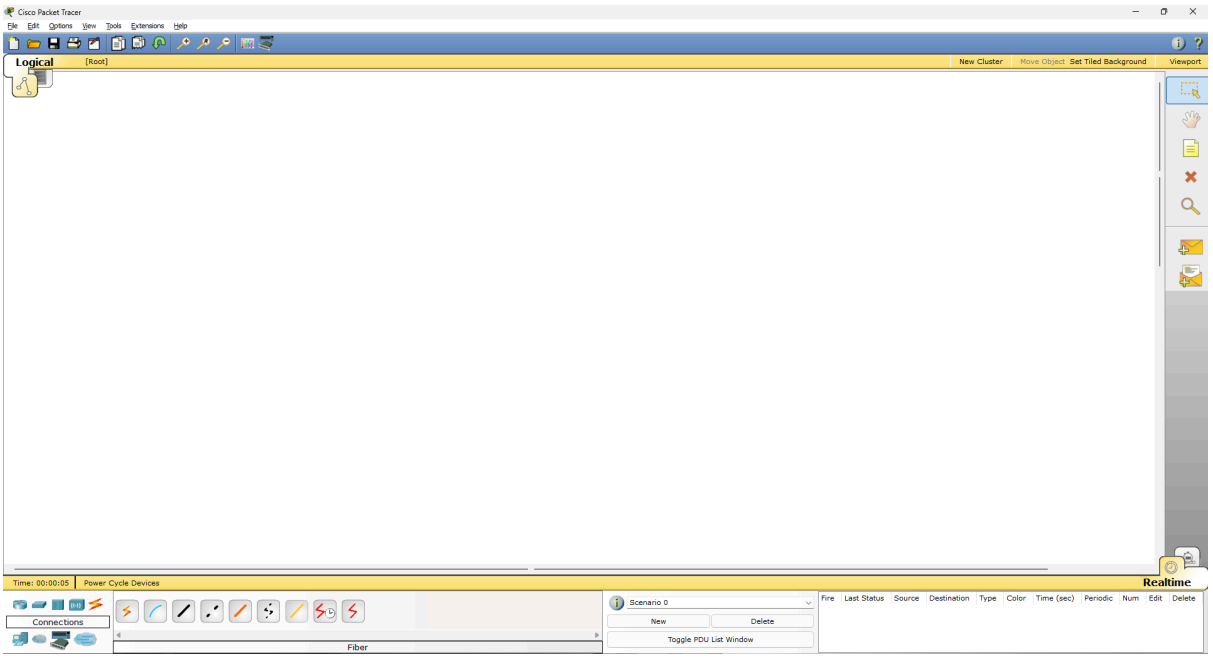
PRACTICALS

Computer Networks Laboratory

Practical 10.

Aim: Configure and analyze Network Address Translation (NAT) protocol in packet tracer to translate private IP addresses to public IP addresses.

Step 1: Open Cisco packet tracer 5.1 as shown in **FIG 1**

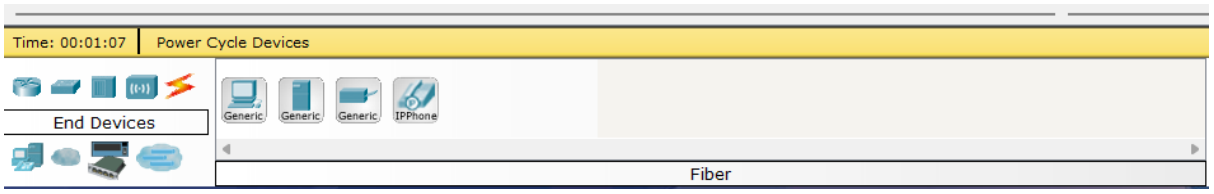


[FIG 1]

Step 2: First we will Understand how to build Topology then we will create various topologies. Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection



[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 3 - PCs
2. 1 - Network Switch
3. 3 - Connecting wires/ cables



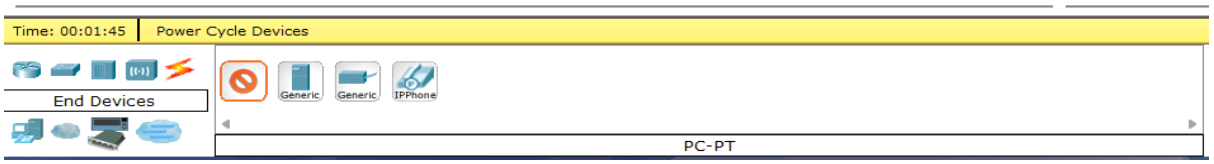


FIG 3

Step 4: Creating Network (Topology) Using selected devices as shown in **FIG 4**

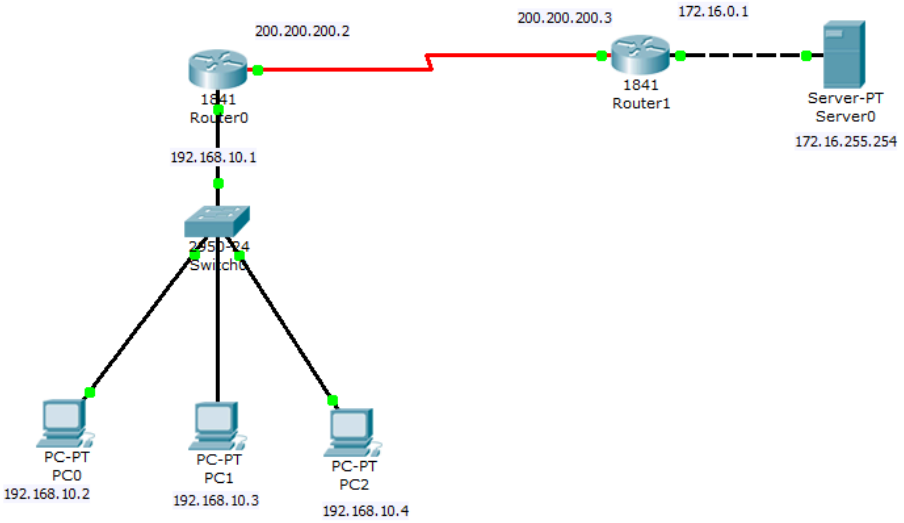


FIG 4

Step 5: Configuring The Setup with Network (static method) Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration

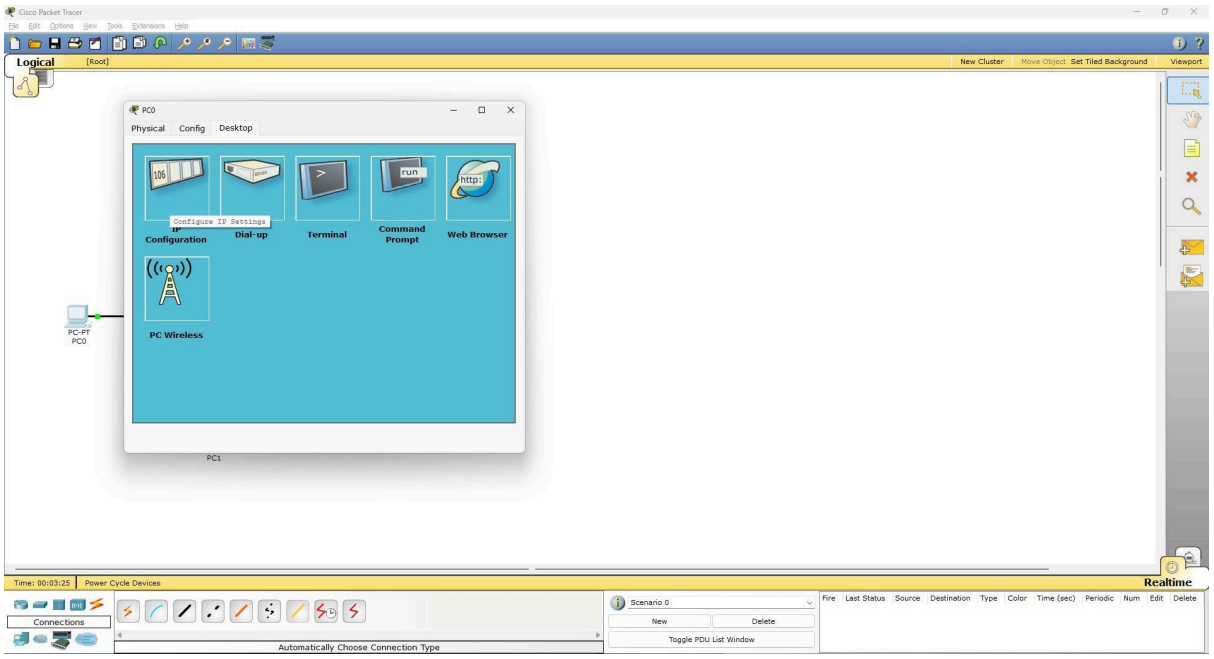


FIG 5

1. Select Or click on Static
2. Enter values as follows
3. Close the dialog box
4. Do the same process for all of the PCs

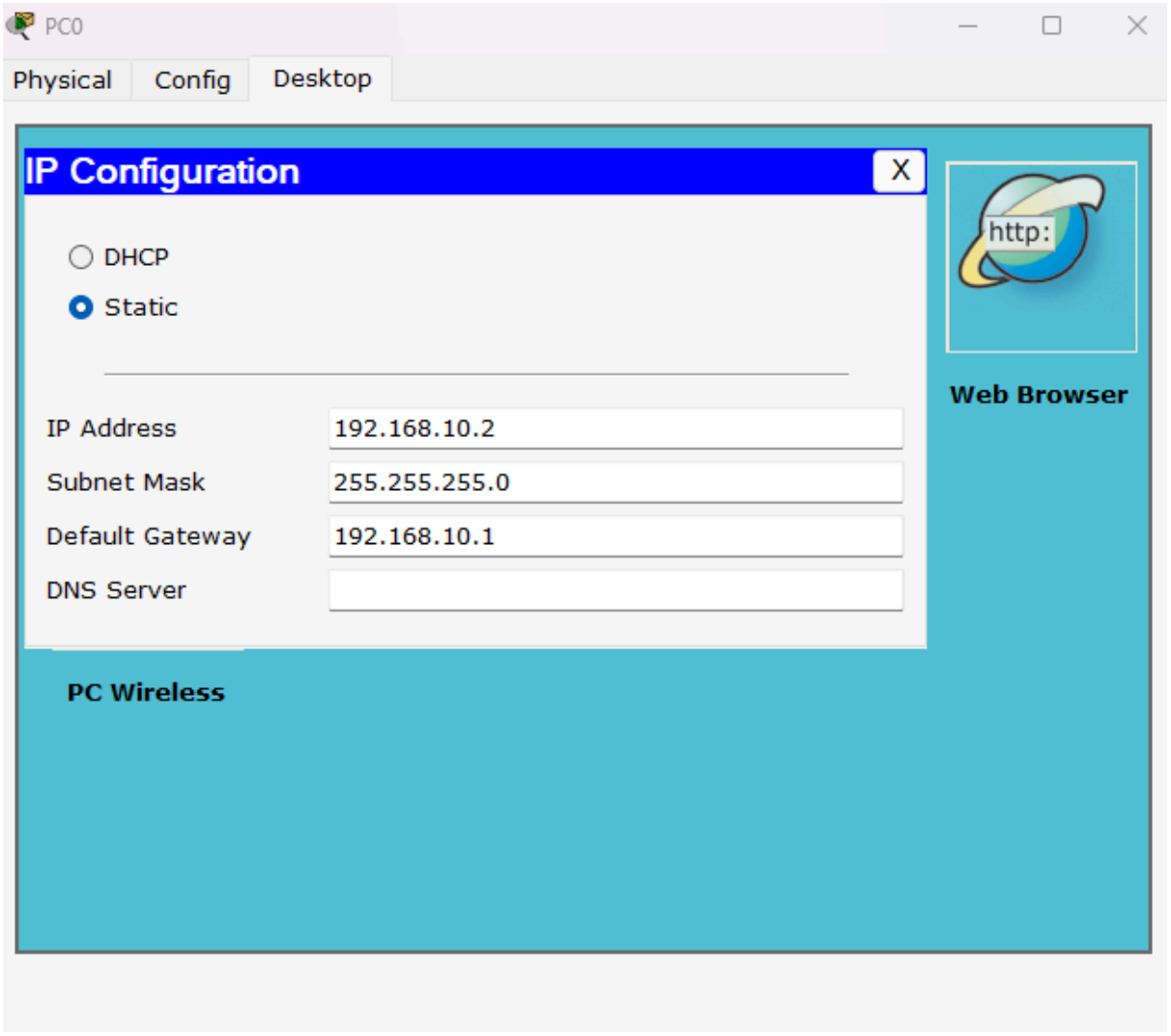


FIG 6

For Server

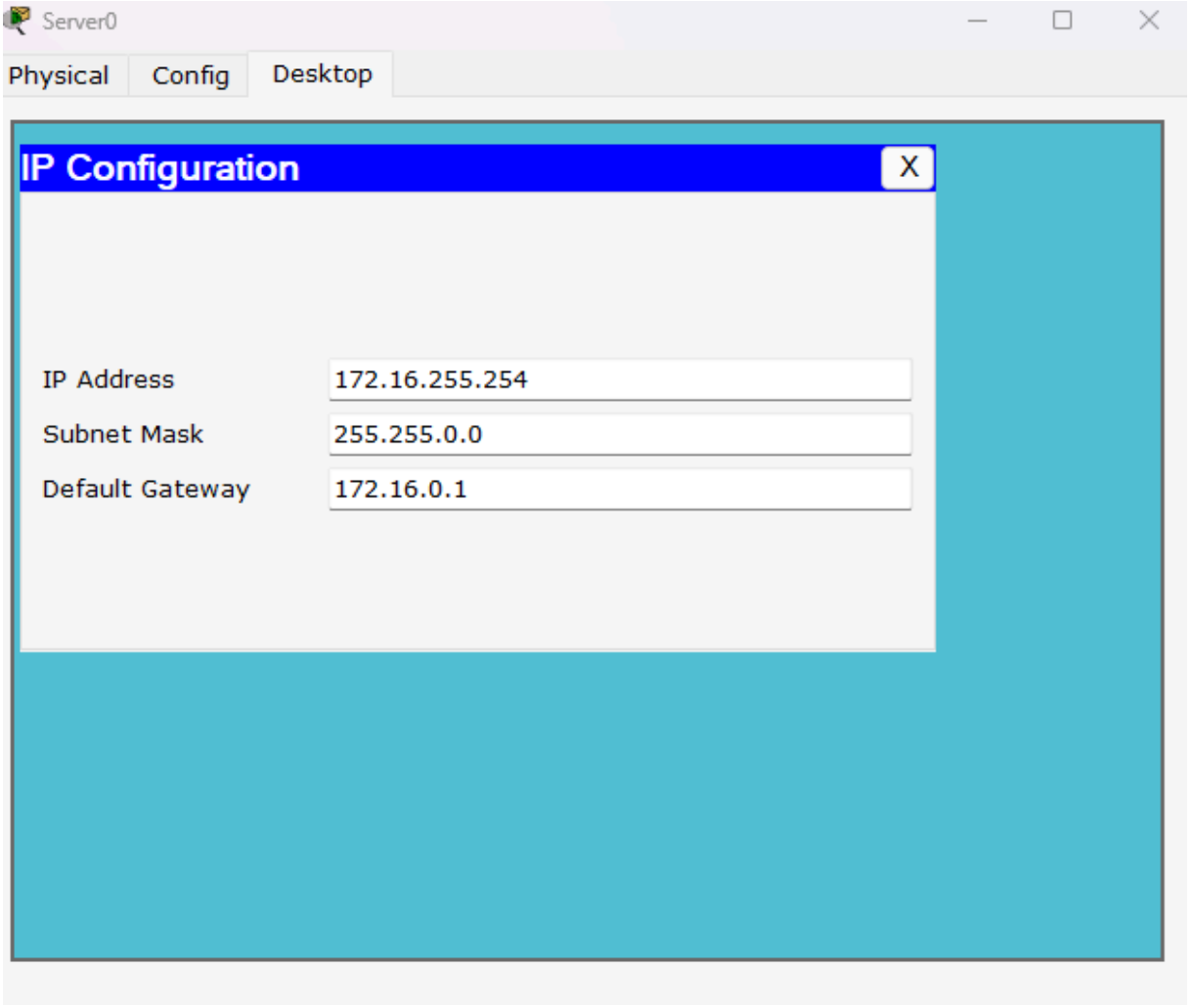


FIG 8

Step 6: Now Click on Router and Go to Config

1. Go To fastEthernet0/0 and serial0/0/0 and add the following Values as shown in **FIG 9.1** and **FIG 9.2**

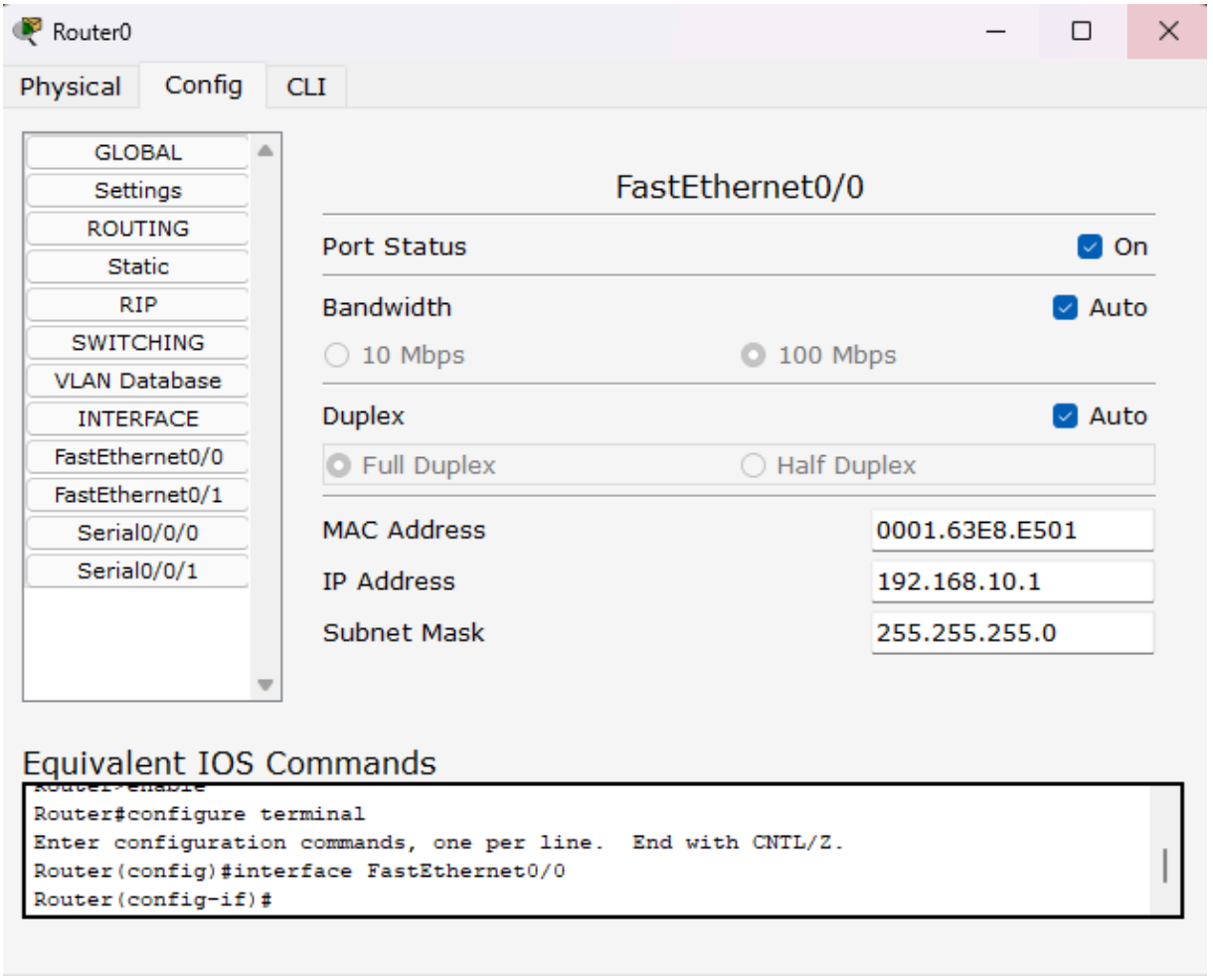


FIG 9.1

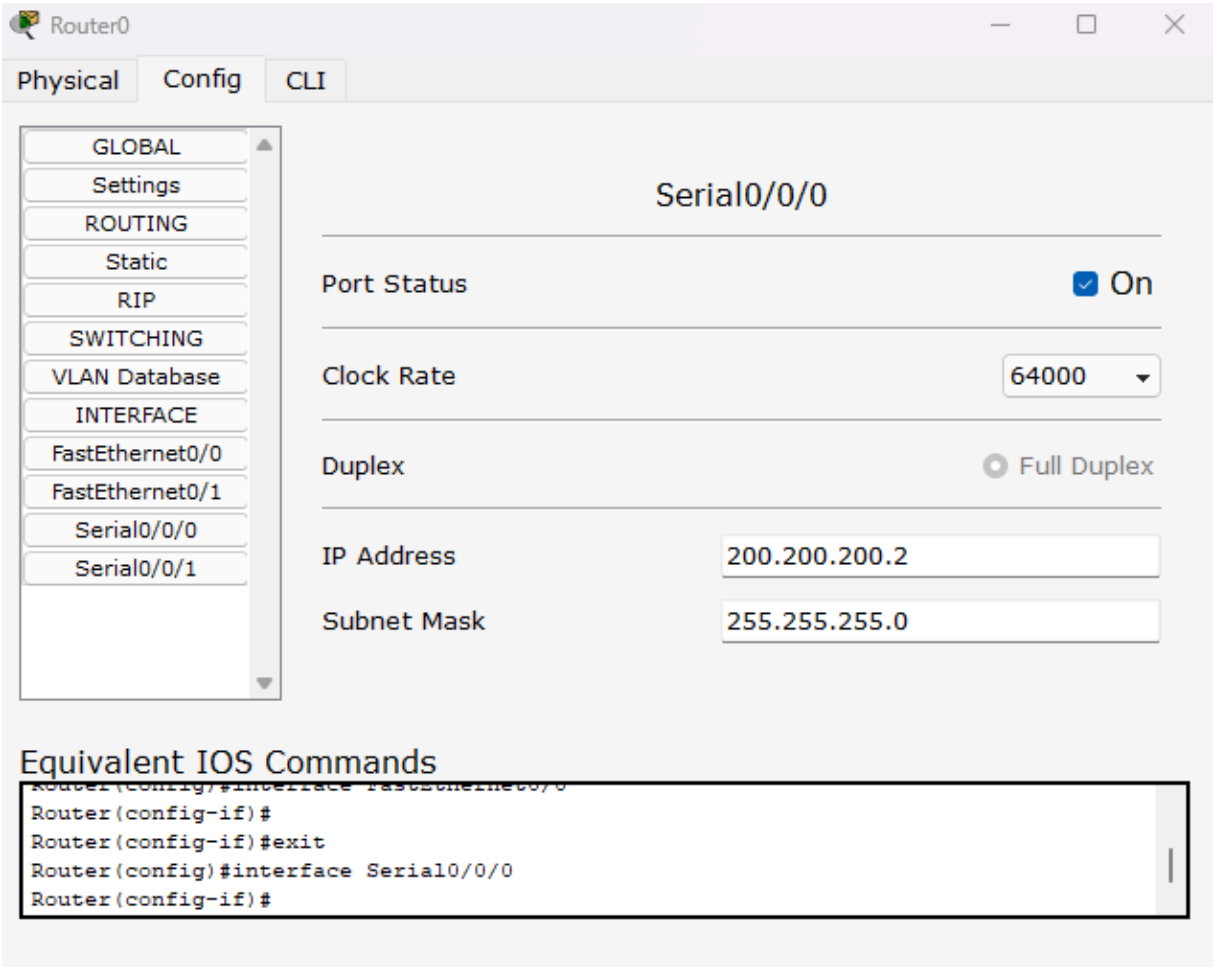

Router0
Physical
Config
CLI
GLOBAL
Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Clock Rate
64000
Duplex
Full Duplex
IP Address
200.200.200.2
Subnet Mask
255.255.255.0
Equivalent IOS Commands
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#

FIG 9.2

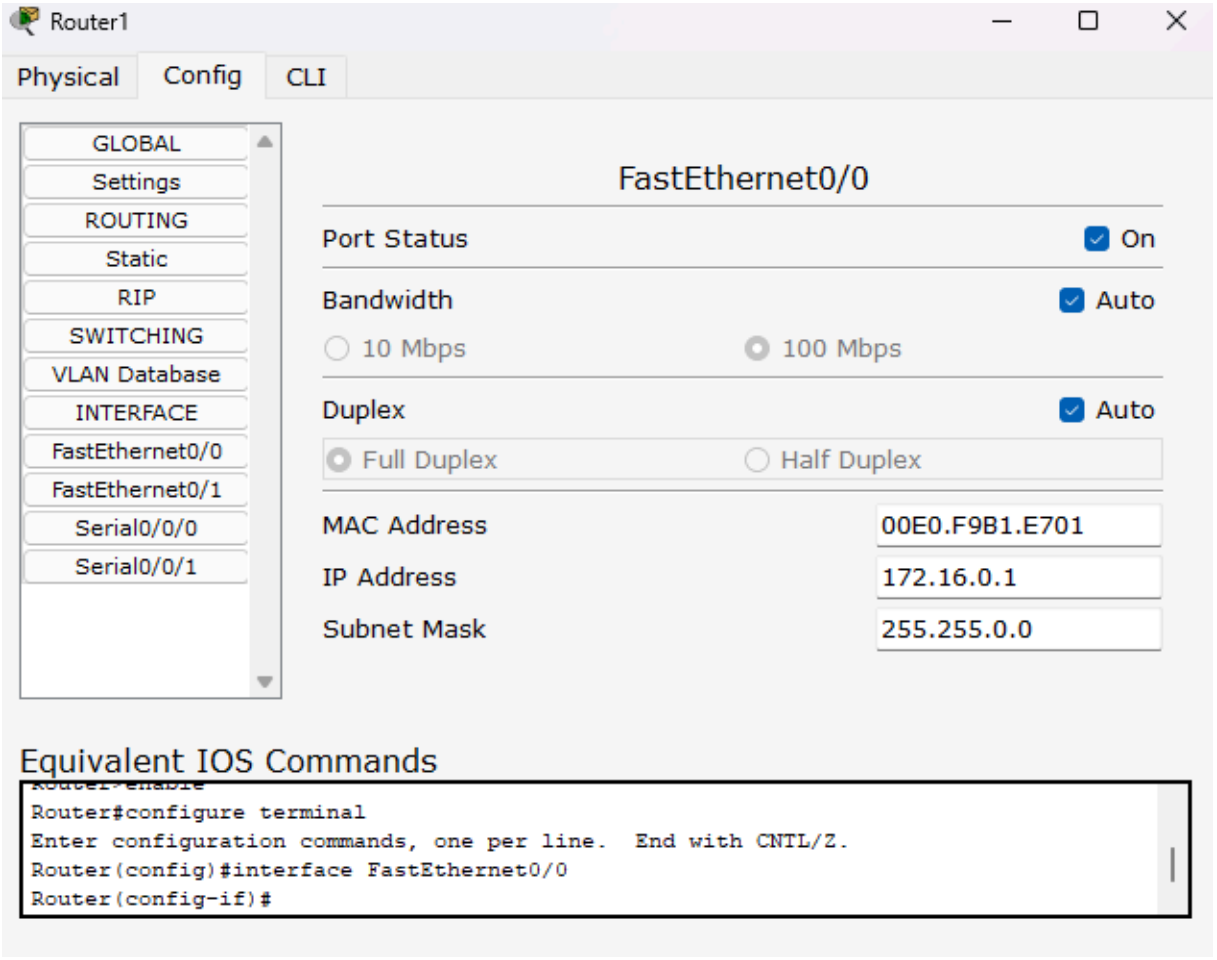

Router1
Physical
Config
CLI
GLOBAL
Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
FastEthernet0/0
Port Status
On
Bandwidth
Auto
10 Mbps
100 Mbps
Duplex
Auto
Full Duplex
Half Duplex
MAC Address
00E0.F9B1.E701
IP Address
172.16.0.1
Subnet Mask
255.255.0.0
Equivalent IOS Commands
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#

FIG 9.3

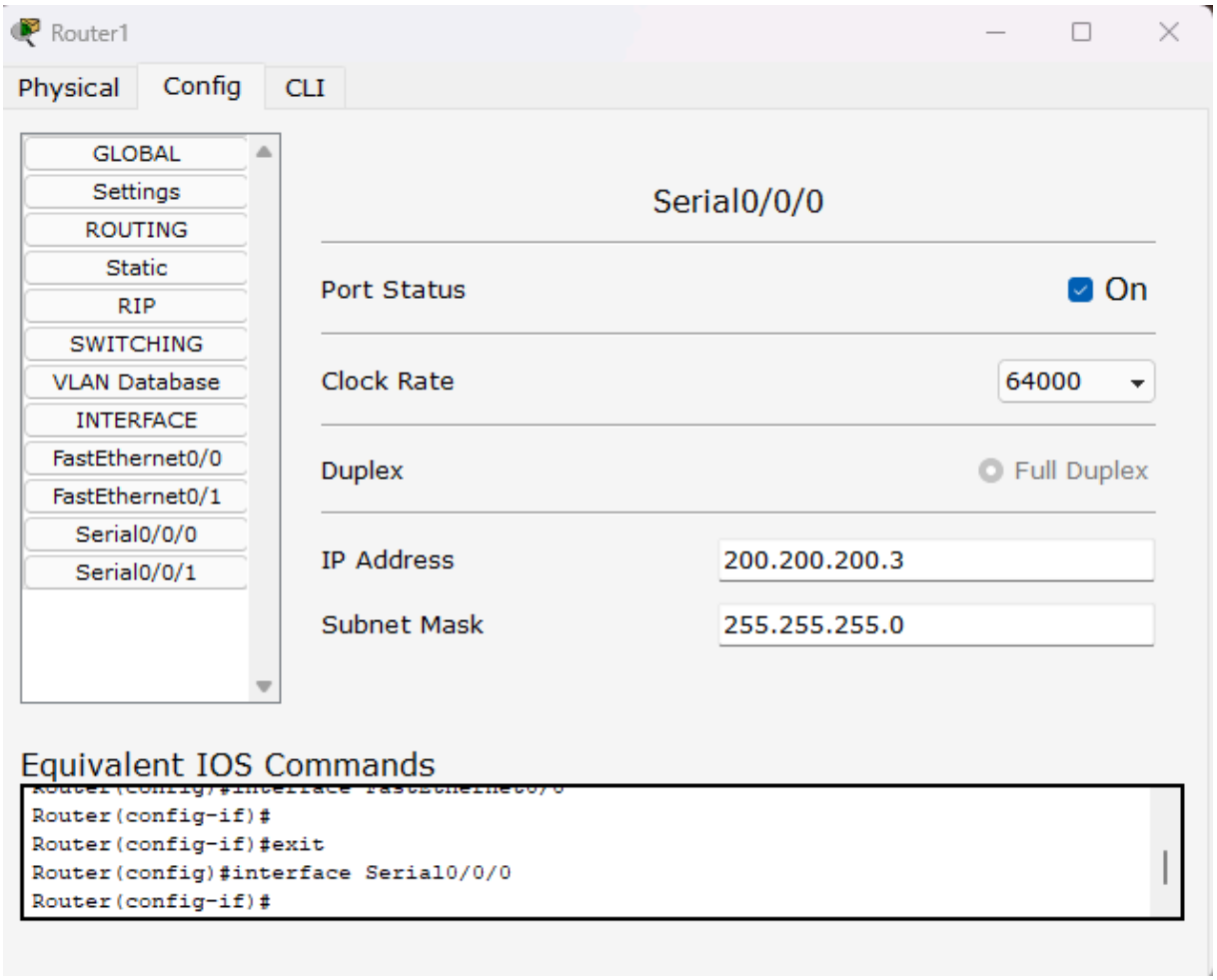


FiG 9.4

Step 6:- Now open CLI of ROUTER 1 and Run the Following Commands

enable
Configure terminal
Interface serial 0/0/0
Ip address 200.200.200.3 255.255.255.0
Clock rate 6400
Exit
Ip route 0.0.0.0 0.0.0.0 serial 0/0/0
Exit

Enable
Configure terminal
Ip nat inside source static 172.16.255.254 200.200.200.3
Exit

Configure terminal

Interface fastEthernet0/0
Ip nat inside
Interface serial 0/0/0
Ip nat outside
Exit

Step 8:- open PC and go to desktop and click on browser and write 200.200.200.3 and click on GO if you get response as same as server's response (as shown in **FIG 10**)

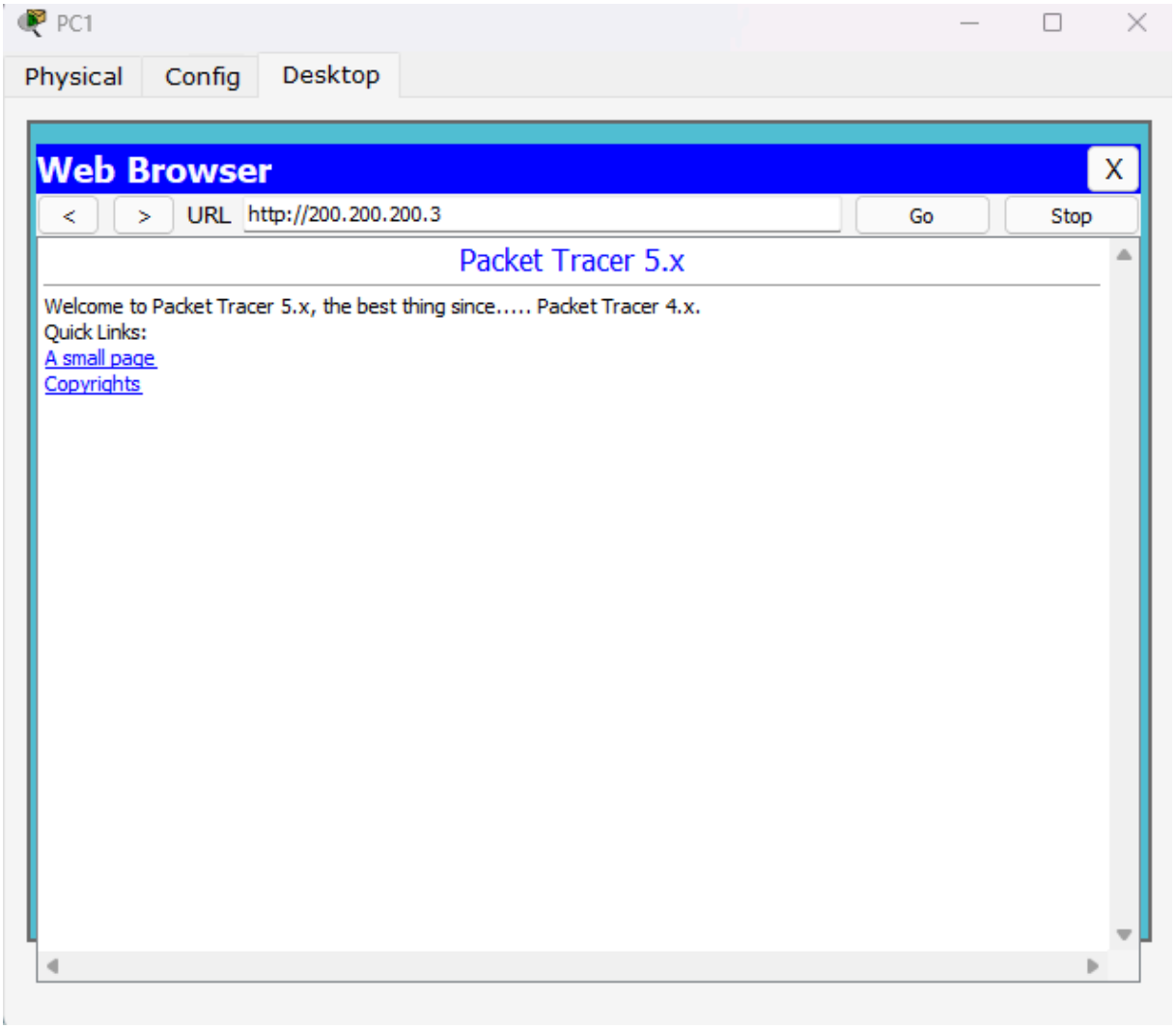


FIG 10

Now Go To Router 0’s CLI and write the following commands

```
enable
config terminal
ip nat inside source list 1 interface serial 0/0/0
access-list 1 permit 192.168.10.0 0.0.0.255
interface fastEthernet0/0
ip nat inside
interface serial0/0/0
ip nat outside
exit
```